TYLER HENDERSON

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Relocating to Montréal | Available August 1st, 2026

PROFESSIONAL SUMMARY

Fenestration technical leader specializing in NFRC/CSA A440.2 compliance, thermal simulation, and certification strategy. ISO/IEC 17065 QMS programs, EPD/LCA service lines, and automation that cuts simulation project turnaround from weeks to days. Active in NFRC's standards process as an ANS Standards Committee member and Task Group participant.

CORE COMPETENCIES

- NFRC / CSA A440.2 Simulation & Certification Strategy
- Compliance Program & ISO/IEC 17065 QMS Development
- EPD / LCA Program Development & Delivery
- Simulation Workflow Automation & Throughput Optimization
- Standards Development & Technical Committee Participation

TECHNICAL SKILLS

Engineering Simulation: LBNL THERM & WINDOW (Advanced), Flixo, Physibel, NFRC/CSA A440.2 & Passive House methodologies
Compliance & Sustainability: ISO/IEC 17065 QMS, OneClick LCA (EPD generation)
Automation & Data: Python, SQL/ODBC, proprietary binary file parsing

PROFESSIONAL DEVELOPMENT & STANDARDS INVOLVEMENT

- Member, NFRC ANS Standards Committee (2026-Present)
- Presenter, NFRC Simulators Task Group (March 2026). Presented independent technical research on two-box spacer modeling methodology and validation

TECHNICAL ACHIEVEMENTS & PUBLICATIONS

Validation of Two-Box Spacer Modeling Across Spacer Widths for Fenestration Thermal Analysis - Independent Technical Report (February 2026)

- Authored an independent technical report evaluating two two-box spacer calibration strategies, per-width calibration and single nominal-width calibration, against detailed spacer models using LBNL THERM and NFRC 100/200/500 methodology.
- Ran 900 full-frame simulations across 3 representative spacer families (composite single seal, metal single seal, metal double seal) and 25 discrete spacer widths from 8-20 mm.
- Found that width-calibrated two-box models reproduce detailed spacer performance with high fidelity, holding full-window U-factor deviations to within approximately 0.008 $W/m^2 \cdot K$ across the full width range.
- Demonstrated that models calibrated at a single nominal width are not generally transferable across spacer widths without recalibration, particularly for spacers with moderate-to-high effective thermal conductivity.

PROFESSIONAL EXPERIENCE

Technical Solutions Lead (R&D & Strategy) | Layton Consulting Ltd., Surrey, BC

January 2024 – December 2025

- Compliance & QMS: Engaged the Standards Council of Canada (SCC) to initiate ISO/IEC 17065 accreditation; performed gap analysis against existing organizational practices, designed and implemented QMS structure, and issued recommendations for remaining accreditation requirements, positioning the firm for expanded certification-body eligibility.
- EPD/LCA Program: Launched a new EPD service line from the ground up and led a multi-product-line engagement using OneClick LCA, delivering 6 EPDs for a major Ontario-based aluminum window manufacturer, opening a new revenue line for the firm.
- Simulation Database Platform: Architected a web-based simulation database (Next.js/Supabase) cataloguing all historical project simulations, with automatic integration into calculation spreadsheets and reporting workflows.
- Simulation SOPs: Defined and implemented Standard Operating Procedures for NFRC-compliant simulation methodology on complex glazing products, ensuring full traceability across projects.
- Workflow Automation: Built Python-based automation tools that cut project turnaround by an order of magnitude, shrinking multi-week simulation engagements to days, and multi-day engagements to hours, unlocking high-volume certification revenue previously inaccessible to the firm.
- Strategic Feasibility: Led technical feasibility analysis for long-term initiatives, including LCA/EPD implementation and data governance/privacy risk.

Fabrication Process Specialist & CNC Lead | Glastech Glazing Contractors Ltd., Surrey, BC

August 2023 - January 2024

- Programmed and optimized RhinoFab1100 and Emmegisoft 3-axis CNC workflows for precision curtain wall fabrication.
- Redesigned warehouse layout and production flow to minimize material handling and increase throughput.
- Developed training protocols for staff on CNC operation and automated fabrication logic.

TECHNICAL PORTFOLIO

- NFRC-101-Materials (github.com/hendersontylerjohn/NFRC-101-Materials) - Automated pipeline that monitors, parses, and validates NFRC 101 material property tables (Appendices A, B, C) from the published PDF, syncing them into structured, versioned JSON via a scheduled GitHub Actions workflow.
- Thm-Reader (github.com/hendersontylerjohn/Thm-Reader) - Rust library for reading and writing THERM 7.x binary .thm files, exposing typed access to header, polygon, and IGU/glazing data plus the material, boundary-condition, and gas libraries, with round-trip byte-accurate serialization and JSON export.

EDUCATION & CERTIFICATIONS

- NFRC LEAFF Certified Simulator - National Fenestration Rating Council (2026)
- Phius Blue Path Certified Simulator - Passive House Institute US (Phius)
- University of British Columbia Okanagan – Coursework towards B.A.Sc, Electrical Engineering, September 2021 - May 2023; including electromagnetism & optics, circuit design, embedded systems, and SolidWorks modeling.